Transfer function: closed loop gain

$$A_{CL} = \frac{V_{out}}{V_{in}}$$

Feedback factor (β): fraction of output signal is fed back into input

loop gain (T): Gain a signal encounters as it travels around feedback loop

$T = A_{OL} \times \beta$ A_{OL} = open loop gain of opamp

$$A_{CL} = \frac{A_{OL}}{1 + A_{OL} \cdot \beta}$$